

Rushiraj Gadhvi

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🎓 [Google Scholar](#)

Technical Skills

Programming Languages: Proficient with Python (PyTorch, gym, transformers, spaCy, NLTK), C++, Dart, SQL
Web/App Frameworks: Flutter, Django, Next.js, Vue.js, Tailwind, FastAPI
Other: Docker, Bash, Git, Linux, GCP, Scientific Writing, Electronics Prototyping (RPI, Arduino, PCB Design)

Experience

RESEARCH COLLABORATOR @HCI Engineering Group, MIT

- Collaborated with [Prof. Stefanie Mueller's](#) group on FilaTwin Project (3D preview system for multi-material 3D printing).

Aug'25 – Mar'26

RESEARCH INTERN @MuRAL Lab, Plaksha University

- Developed energy-efficient navigation system for autonomous surface vehicles as part of the Khalasi Project under [Prof. Sandeep Manjanna](#).
- Built simulation frameworks to evaluate energy-efficient trajectory planning in dynamic aquatic environments.

Jun – Aug'25

FOUNDER @Housify (Techsurf Solutions)

- Designed and launched a fin-tech platform to streamline residential society administration and increase transparency.

Jan'25 – Jan'26

RESEARCH INTERN @HTI Lab, Plaksha University

- Leading research on PosePilot (real-time posture correction system) and AdaptAI (multimodal productivity system) under [Prof. Siddharth](#).
- Published *two papers* in *top conferences including CHI, 2025*.

Jan'24 – Feb'25

CONTRIBUTOR @Google Summer of Code (GSoC) 2024

- Improved the visualization tools for quantum circuits in the open-source QuTiP library ([project webpage](#)), widely used in quantum information research. Mentor: Dr. Boxi Li, [Dr. Neill Lambert](#), [Dr. Alex Pitchford](#)

Jun – Aug'24

DATA SCIENCE INTERN @BlinkX by JM Financial, Mumbai

- Designed and developed an end-to-end internal tool for processing application feature-based opinions from customer reviews to support strategic decision-making using LLMs, RAG and NLP-based clustering techniques.

Jun – Jul'24

RESEARCH INTERN @Physical Research Laboratory (PRL), Ahmedabad

- Fine-tuned models for spectral data composition analysis using statistical techniques and developed a Python based spectrum visualization platform that streamlined data exploration and reduced analysis time for large datasets. Mentored by [Dr. Arvind Singh](#).

Jun – Jul 2023

Education

Plaksha University

B.Tech in Computer Science & Artificial Intelligence (CSAI)

Mohali, India

2021 – 2025

Edunova High School

11th/12th (A-Stream: Math, Physics, Chemistry, Computer)

Ahmedabad, India

2019 – 2021

Achievements

- AlgoVerse's AI Safety Research Fellowship 2026** — One amongst 30+ selected globally
- GSoC (Google Summer of Code) 2024 @numFOCUS** — amongst 1,220 selected globally, out of 43k+ applicants
- AWS AI & ML Scholar Advance 2024** — One amongst 500 selected globally
- AWS AI & ML Scholar 2023** — One amongst 2k selected globally, out of 70k+ applicants

- **UN Millennium Fellowship 2022** — One amongst 3k selected globally, out of 31k+ applicants
- **1st rank** amongst 190+ teams at **IIT Bombay E-Cell BnB Event 2022**
- **Student Startup and Innovation Policy (SSIP) 2020, Gujarat** amongst 200+ applicants

Projects

How Language Models Choose Sides (w/ Algoverse)

Present [Working]

Internal Representations of Instruction Hierarchy

- Studied how instruction-tuned LLMs arbitrate **system-vs-user instruction conflicts** — critical for preventing **prompt injection attacks** and ensuring models follow **developer-intended behaviour**.
- Introduced a **41-scenario conflict benchmark**; found conflict outcomes to be **linearly decodable** from residual-stream activations at **0.97 balanced accuracy**.
- Causal intervention via **logistic-regression directions** steered system compliance from **13.2% → 53.0%**.

FilaTwin (w/ MIT HCI Engineering Group, under review)

Aug 2025 – Mar 2026

A Rotating Digital Preview for Multi-Material 3D Printing

- Current slicing tools only visualize toolpaths and cannot simulate how filaments actually appear when printed; FilaTwin extracts **material properties** from **reference photos of prior prints** and synthesizes them into **procedural shaders** for live, rotatable 3D preview.
- Designed a **VLM-driven pipeline with iterative refinement** and **natural-language feedback** support, enabling appearance-aware previewing across single- and multi-filament prints, including gradient, color-shifting, and transparent filaments.
- Evaluated against VisiPrint in a **12-participant user study**; achieved **~2× faster task completion**, significantly higher closeness-to-real-print ratings, and was **preferred by all participants**.

Khalasi [✉](#) (under review)

Nov 2024 – Sep 2025

Energy-Efficient Navigation for Surface Vehicles in Vortical Flow Fields

- Developed a **continuous control RL framework** able to optimize energy consumption in dynamic vortical environments by introducing "flow-surfing" behavior. Achieving **30-50% reduction in energy consumption**.
- Architected a hybrid **Soft Actor-Critic (SAC)** pipeline dealing with partial observability; integrated **Gaussian Process Regression (GPR)** for local flow reconstruction.
- Demonstrated **robust zero-shot generalization by training on variety of custom environments** and deploying on **real-world ocean current data and complex synthetic gyre formations** without fine-tuning.

AdaptAI [✉](#) (CHI'25)

Nov 2024 – Feb 2025

A Personalized Solution to Sense Your Stress, Fix Your Mess, and Boost Productivity

- AdaptAI is a multimodal AI system that integrates real-time data from egocentric vision, audio, and physiological signals to enhance productivity and well-being by tailoring interventions to individual needs.
- Developed a **tone-adaptive LLM conversational agent that conditions its linguistic style on real-time physiological stress signals**, enabling supportive, context-sensitive dialogue grounded in HRV-based user state modeling.
- Conducted a **15-participant pilot study**, demonstrating **significant gains in task throughput and user satisfaction**.

Publications

How Language Models Choose Sides: Internal Representations of Instruction Hierarchy

Enrique Balp-Straffon, Chih-Hao Hsu, **Rushiraj Gadhvi**

Sunishchal Dev, Callum Stuart McDougall, Anusha Mujumdar

MechInterp

ICML, 2026

[OpenReview](#) [✉](#)

FilaTwin: A Rotating Digital Preview for Multi-Material 3D Printing

Rushiraj Gadhvi, Maxine Perroni-Scharf, Stefanie Mueller

Under Review

Khalasi: Energy-Efficient Navigation for ASVs in Turbulent Vortical Fields

Rushiraj Gadhvi, Sandeep Manjanna

Under Review

[arXiv](#) [✉](#)

PosePilot: Edge-AI Solution for Posture Correction in Physical Exercises

Rushiraj Gadhvi, Priyansh Desai, Siddharth

IbPRIA, 2025

[DOI](#) [✉](#) [arXiv](#) [✉](#)

AdaptAI: A Personalized Solution to Sense Your Stress, Fix Your Mess, and Boost Productivity

CHI, 2025

[DOI](#) [✉](#) [arXiv](#) [✉](#)

Certifications

- Bluedot's Technical AI Safety ([Certificate ↗](#))
- Introduction to Cooperative AI by CAIF([Certificate ↗](#))
- Bluedot's AGI Strategy ([Certificate ↗](#))
- Introduction to fundamentals of AI Safety (ENAIIS) ([Certificate ↗](#))
- Udacity's Nanodegree in AWS Machine Learning Fundamentals ([Certificate ↗](#))
- Udacity's Nanodegree in AI Programming with Python ([Certificate ↗](#))